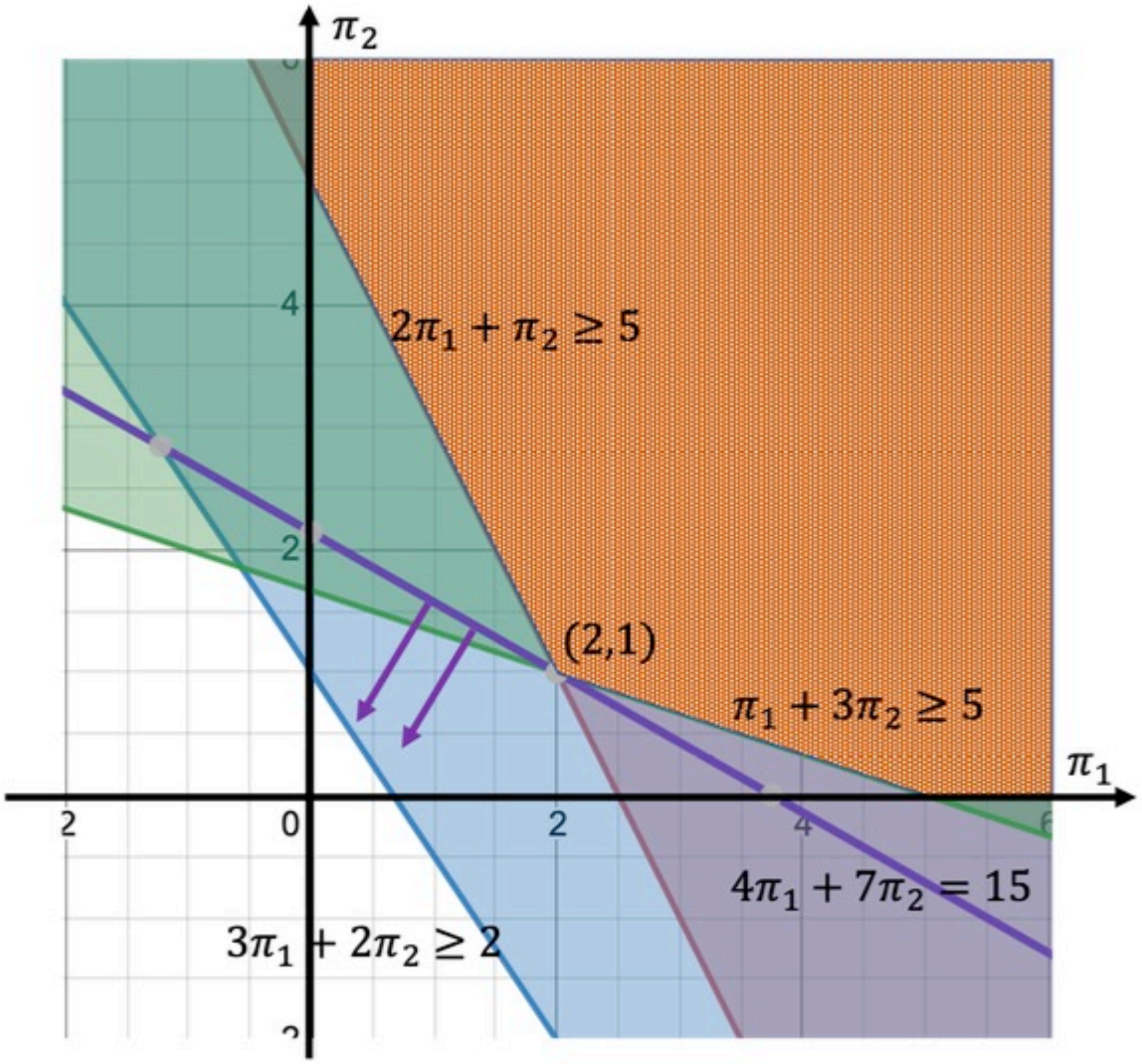
Assignment 4 Solution

1.

(a) Dual Problem:

$$\begin{aligned} \min \quad & 4y_1 + 7y_2 \\ \text{s.t.} \quad & 2y_1 + y_2 \geq 5 \\ & 3y_1 + 2y_2 \geq 2 \\ & y_1 + 3y_2 \geq 5 \\ & y_1, y_2 \geq 0 \end{aligned}$$

(b)



Optimal value = [2,1]

Optimal value = 15

(c)
Based on complementatry slackness condition theory, we have

$$
\begin{aligned}
(2y_1 + y_2 - 5)x_1 &= 0 \\
(3y_1 + 2y_2 - 2)x_2 &= 0 \\
(y_1 + 3y_2 - 5)x_3 &= 0 \\
(2x_1 + 3x_2 + x_3 - 4)y_1 &= 0 \\
(x_1 + 2x_2 + 3x_3 - 7)y_2 &= 0
\end{aligned}
$$

The optimal solution is $[1, 0, 2]$.
The optimal value is 15.

2.

(a) Let x_1, x_2, x_3, x_4 represent the quantities of Bread, Milk, Fish, and Potato, respectively, that should be consumed.
The problem can be formed by

$$
\begin{array}{lll}
\min & 2x_1 + 3.5x_2 + 8x_3 + 1.5x_4 & \\
\text{s.t.} & 4x_1 + 6x_2 + 20x_3 + 1x_4 & \geq 25 \\
& 8x_1 + 12x_2 + 0x_3 + 30x_4 & \geq 40 \\
& 130x_1 + 120x_2 + 150x_3 + 70x_4 & \geq 400 \\
& x_1, x_2, x_3, x_4 \geq 0 &
\end{array}
$$

(b)

```
f2 = [2,3.5,8,1.5];
A2 = [4,6,20,1;8,12,0,30;130,120,150,70];
b2 = [25,40,400];

lb2 = [0,0,0,0];
[x2, fval2] = linprog(f2, -A2, -b2, [], [], lb2, [], options);
maxValue = fval2;
disp('Solution x:');
disp(x2);
disp('Maximum value of the objective function:');
disp(maxValue);
```

```
Solution x:
    1.5500
         0
    0.8940
    0.9200

Maximum value of the objective function:
   11.6320
```

(c)

The dual problem seeks to maximize the total value derived from selling the minimum required nutrients, subject to the condition that buying the nutrients from the pharmaceutical company doesn't exceed the cost of obtaining them through the different foods.

$$
\begin{aligned}
\max \quad & 25y_1 + 40y_2 + 400y_3 & \\
\text{s.t.} \quad & 4y_1 + 8y_2 + 130y_3 & \leq 2 \\
& 6y_1 + 12y_2 + 120y_3 & \leq 3.5 \\
& 20y_1 + 0y_2 + 150y_3 & \leq 8 \\
& 1y_1 + 30y_2 + 70y_3 & \leq 1.5 \\
& y_1, y_2, y_3 \geq 0 &
\end{aligned}
$$

(4)

```
f3 = [25,40,400];
A3 = [4,8,130;6,12,120;20,0,150;1,30,70];
b3 = [2,3.5,8,1.5];
lb3 = [0,0,0];
[y3, fval3] = linprog(-f3, A3, b3, [], [], lb3, [], options);
maxValue = -fval3;
disp('Solution y:');
disp(y3);
disp('Maximum value of the objective function:');
disp(maxValue);
```

```
Solution y:
    0.3904
    0.0340
    0.0013

Maximum value of the objective function:
   11.6320
```

3.

First, we will show that it can't both have solutions.
Suppose both (a) and (b) are feasible, then we have

$$Ax \leq b$$
$$y^T A = 0, b^T y < 0, y \geq 0$$

for some x and y.
However, since $y \geq 0$ and $Ax \leq b$, it follows that: $y^T b \geq y^T Ax = 0$
This contradicts the assumption that $b^T y < 0$. So (a) and (b) cannot have solutions simutanously.

Second, we show that if the second system is infeasible, then the first system must be feasible. We consider the following pair of linear optimization problems:

$$\begin{array}{ll} \min & b^\top y \\ \text{s.t.} & A^\top y = 0 \\ & y \geq 0. \end{array}$$

The dual of this problem is

$$\begin{array}{ll} \max & 0 \\ \text{s.t.} & Ax \leq b \end{array}$$

If the second system does not have a solution, then the primal problem can't attain negative objective value. In the meantime, $y = 0$ is always a feasible solution for the primal problem with objective value 0 . Therefore, $y = 0$ must be an optimal solution to the primal problem. Then by the strong duality theorem, the dual problem must also be feasible. Thus, the result is proved.

4.

(a)

Let x_1, x_2, x_3 be the number of times Process 1, Process 2, and Process 3 are run, respectively.
So the problem can be expressed as:

$$\begin{aligned} \max \quad & 200x_1 + 60x_2 + 206x_3 \\ \text{s.t.} \quad & 3x_1 + x_2 + 5x_3 \leq 8000 \\ & 5x_1 + x_2 + 3x_3 \leq 5000 \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Table 1	$\mathbf{C_j}$	**200**	**60**	**206**	**0**	**0**	
$\mathbf{C_b}$	Base	$\mathbf{X_1}$	$\mathbf{X_2}$	$\mathbf{X_3}$	$\mathbf{S_1}$	$\mathbf{S_2}$	$\mathbf{R}$
0	$\mathbf{S_1}$	3	1	5	1	0	8000
0	$\mathbf{S_2}$	5	1	3	0	1	5000
	$\mathbf{Z}$	-200	-60	-206	0	0	0

The pivot element is 5

Table 2	$\mathbf{C_j}$	**200**	**60**	**206**	**0**	**0**	
$\mathbf{C_b}$	Base	$\mathbf{X_1}$	$\mathbf{X_2}$	$\mathbf{X_3}$	$\mathbf{S_1}$	$\mathbf{S_2}$	$\mathbf{R}$
206	$\mathbf{X_3}$	$3/5$	$1/5$	1	$1/5$	0	1600
0	$\mathbf{S_2}$	$16/5$	$2/5$	0	$-3/5$	1	200
	$\mathbf{Z}$	$-382/5$	$-94/5$	0	$206/5$	0	329600

The pivot element is 16/5

Table 3	$\mathbf{C_j}$	**200**	**60**	**206**	**0**	**0**	
$\mathbf{C_b}$	Base	$\mathbf{X_1}$	$\mathbf{X_2}$	$\mathbf{X_3}$	$\mathbf{S_1}$	$\mathbf{S_2}$	$\mathbf{R}$
206	$\mathbf{X_3}$	0	$1/8$	1	$5/16$	$-3/16$	$3125/2$
200	$\mathbf{X_1}$	1	$\mathbf{1/8}$	0	$-3/16$	$5/16$	$125/2$
	$\mathbf{Z}$	0	$-37/4$	0	$215/8$	$191/8$	334375

The pivot element is 1/8

Table 4	$\mathbf{C_j}$	200	60	206	0	0	
$\mathbf{C_b}$	Base	$\mathbf{X}_1$	$\mathbf{X}_2$	$\mathbf{X}_3$	$\mathbf{S}_1$	$\mathbf{S}_2$	$\mathbf{R}$
206	$\mathbf{X}_3$	-1	0	1	$1/2$	$-1/2$	1500
60	$\mathbf{X}_2$	8	1	0	$-3/2$	$5/2$	500
	$\mathbf{Z}$	74	0	0	13	47	339000

The optimal value is 339000, with $x_1 = 0, x_2 = 500, x_3 = 1500$.

(b) Since the selling price for the Gasoline rised, the objective function has changed. Suppose the changing amount is δ,
originally the coefficient $c = [200, 60, 206, 0, 0]$. Now the new coefficient becomes $c + \Delta c$, where $\Delta c = [4\delta, \delta, 3\delta, 0, 0]$.
From the table 4, we know that

$$A_B^{-1} = \begin{pmatrix} 1/2 & -1/2 \\ -3/2 & 5/2 \end{pmatrix}$$

The new coefficient for the basis variable is $c_b' = [206 + 3\delta, 60 + \delta]$.
To make the optimal solution unchanged, we should have the new reduced cost still remains positive.
Thus,

$$c_B A_B^{-1} \geq 0$$

Taking the value above, we have $\delta \geq -47$.

Also, form the table we know that $A_B^{-1} N = [-1, 8]^T$, $c_N' = [200 + 4\delta]$.
We have $c_B' A_B^{-1} N - c_N' \geq 0$.
So we have $\delta \geq -74$.
Thus, if the price fopr selling A rise, the optimal solution will always remain the same.

(c) The new optimization problem becomes

$$\begin{aligned} \max \quad & 200x_1 + 60x_2 + 206x_3 \\ \text{s.t.} \quad & 3x_1 + x_2 + 5x_3 \leq 8000 \\ & 5x_1 + x_2 + 3x_3 \leq 5000 \\ & 4x_1 + 3x_2 + 5x_3 \leq 10000 \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Using MATLAB, we have
Optimal value = 339000
Optimal solution = $[0, 500, 1500]$

```
function [x, fval] = solveLP()
    % Objective function coefficients
    f = [-200; -60; -206];

    A = [3 1 5; 5 1 3; 4 3 5];
    b = [8000; 5000; 10000];

    Aeq = [];
    beq = [];

    lb = [0; 0; 0];

    ub = [Inf; Inf; Inf];

    [x, fval] = linprog(f, A, b, Aeq, beq, lb, ub);

    fval = -fval;
    disp(fval);
    disp(x);
end
```

5.

(a) Let x_i idenote the number of i th insurance product. The linear program is

$$\begin{aligned} \text{maximize} \quad & 500x_1 + 250x_2 + 600x_3 \\ \text{s.t.} \quad & 2x_1 + x_2 + x_3 \leq 240 \\ & 3x_1 + x_2 + 2x_3 \leq 150 \\ & x_1 + 2x_2 + 4x_3 \leq 180 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

(b) From the final tableau, we see that the shadow prices are $\$0, \$140, \$80$ respectively. If the company sells one unit of mortgage, it requires 1 hour for underwriting, 1 hour for administration, and 2 hours for claims. So, the cost to sell 1 unit of mortgage is $140 + 280 = 300$. However, the profit for one unit of mortgage is only $\$250$ per unit. Therefore, the company cannot make any net profit by selling mortgage.

(c) We have $B = \{4, 1, 3\}$, and we need to determine the range of λ such that

$$\bar{b} + \lambda A_B^{-1} e_3 \geq 0$$

The last three columns in the final simplex tableau corresponds to $A_B^{-1} D = A_B^{-1} I = A_B^{-1}$ (Dcontains the columns of the coefficient matrix A corresponding to slack variables, which forms an identity matrix). Therefore, we can directly read A_B^{-1} from the last three columns of the final tableau,

$$A_B^{-1} = \begin{pmatrix} 1 & -0.7 & 0.1 \\ 0 & 0.4 & -0.2 \\ 0 & -0.1 & 0.3 \end{pmatrix}$$

It is then straightforward to verify that we must have $-130 \leq \lambda \leq 120$, and therefore $b_3 \in [50, 300]$.

More comments on reading A_B^{-1} : What if there are no columns that can form an identity matrix in A ? In this case, if there are columns that forms a diagonal matrix, we can still use the final tableau to get A_B^{-1} without explicitly computing the inverse. For instance, let us assume

$$A = \begin{pmatrix} 1 & 2 & 1 & 0 & 0 \\ 3 & 0 & 0 & 2 & 0 \\ 5 & 5 & 0 & 0 & 3 \end{pmatrix}$$

We can see $[A_3, A_4, A_5]$ forms a diagonal matrix. Define $D := [A_3, A_4, A_5]$. Suppose we read $S := A_B^{-1} D$ from the the corresponding columns of the final tableau. Then, we can calculate A_B^{-1} as

$$A_B^{-1} = S D^{-1}$$

This approach relies on the fact that the inverse of the diagonal matrix D is directly computable.

(d) The optimal basis remains the same if and only if new reduced cost is nonnegative, i.e., $\bar{c}_N - \lambda A_N^T \left(A_B^{-1}\right)^T e_1 \geq 0$

$$\begin{cases} 50 \geq 0 \\ 140 - 0.4\lambda \geq 0 \\ 80 + 0.2\lambda \geq 0 \end{cases}$$

We can find $A_B^{-1} A_N$ from the final tableau, and it is therefore straightforward to verify that $-400 \leq \lambda \leq 350$, and therefore that $c_1 \subset [150, 900]$.